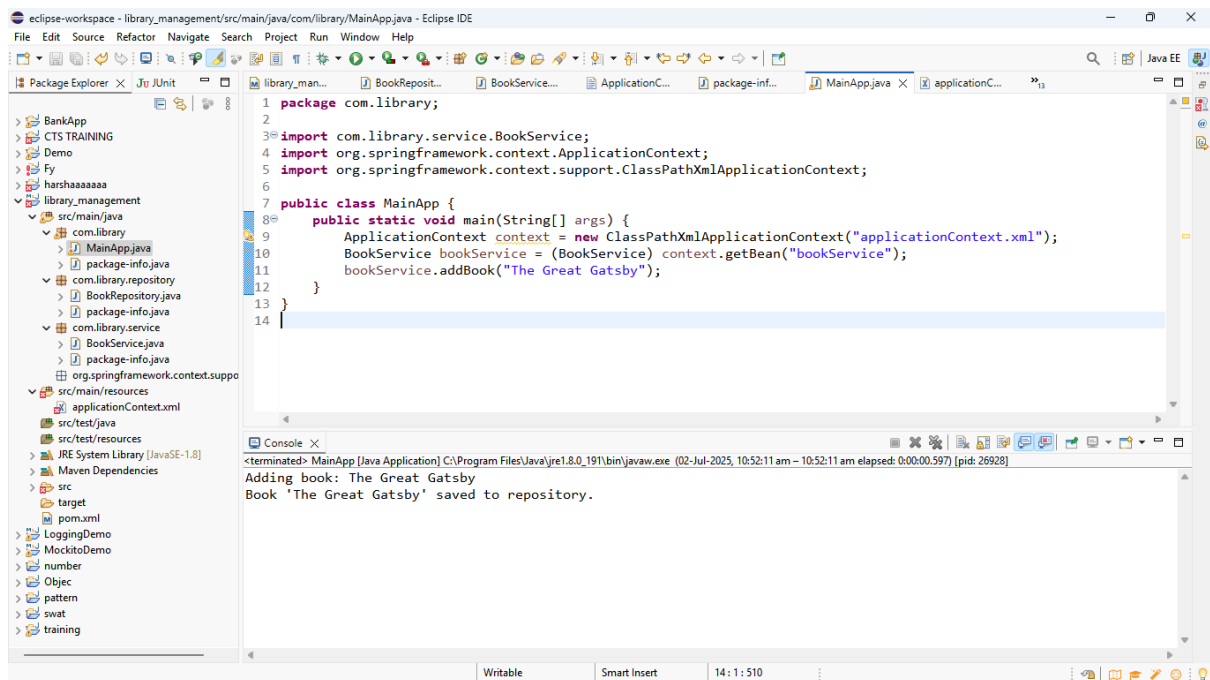
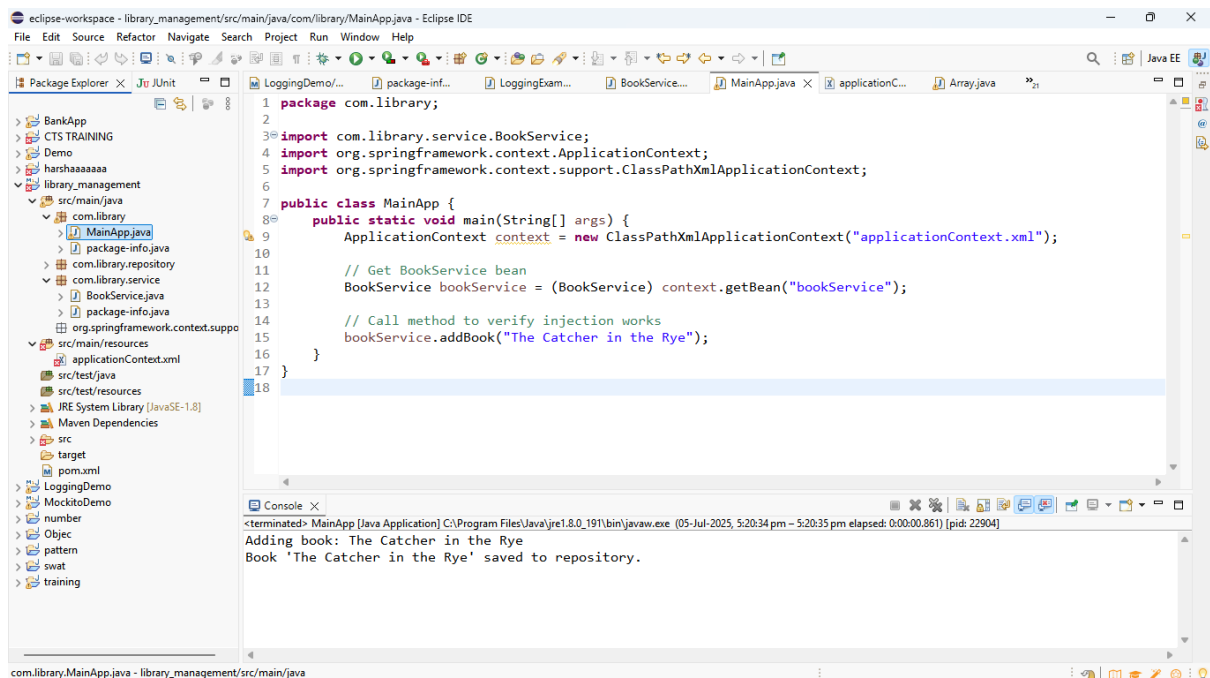


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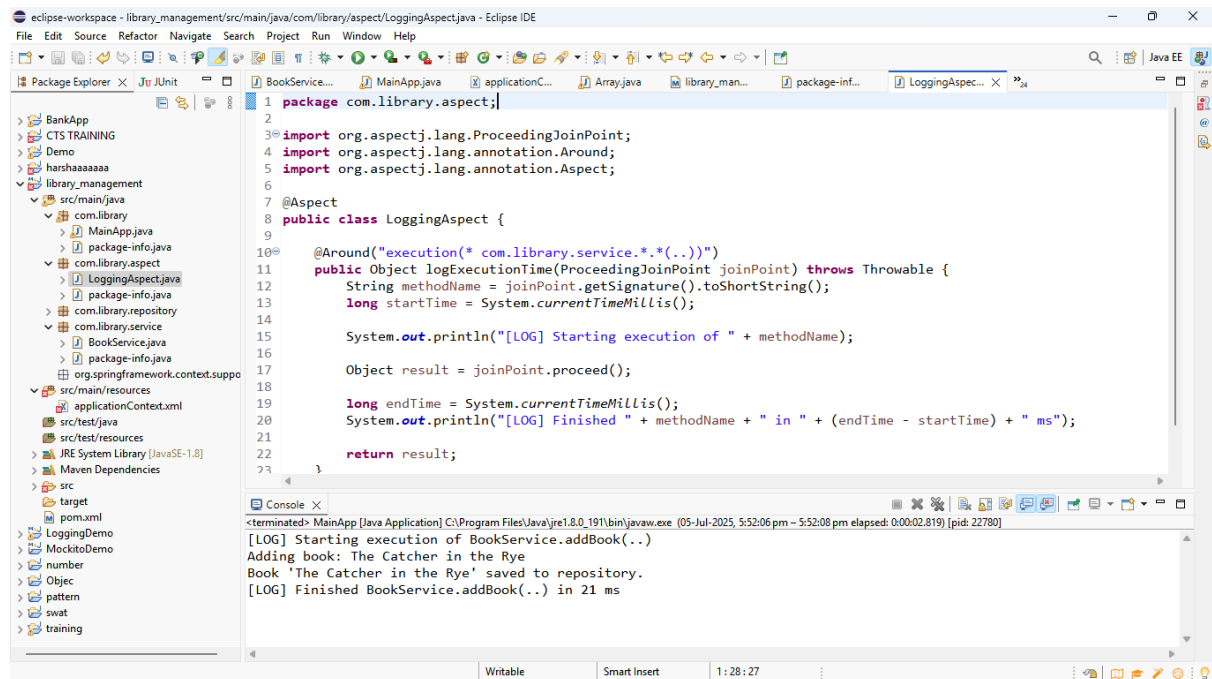
Exercise 1: Configuring a Basic Spring Application



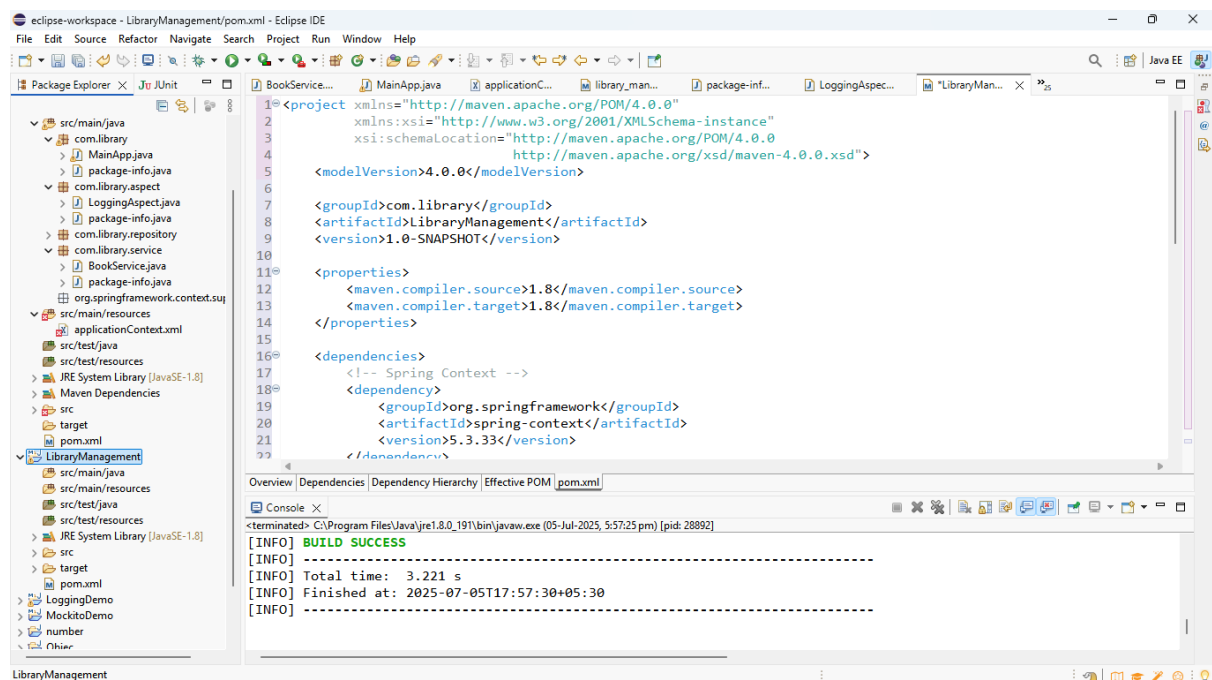
Exercise 2: Implementing Dependency Injection



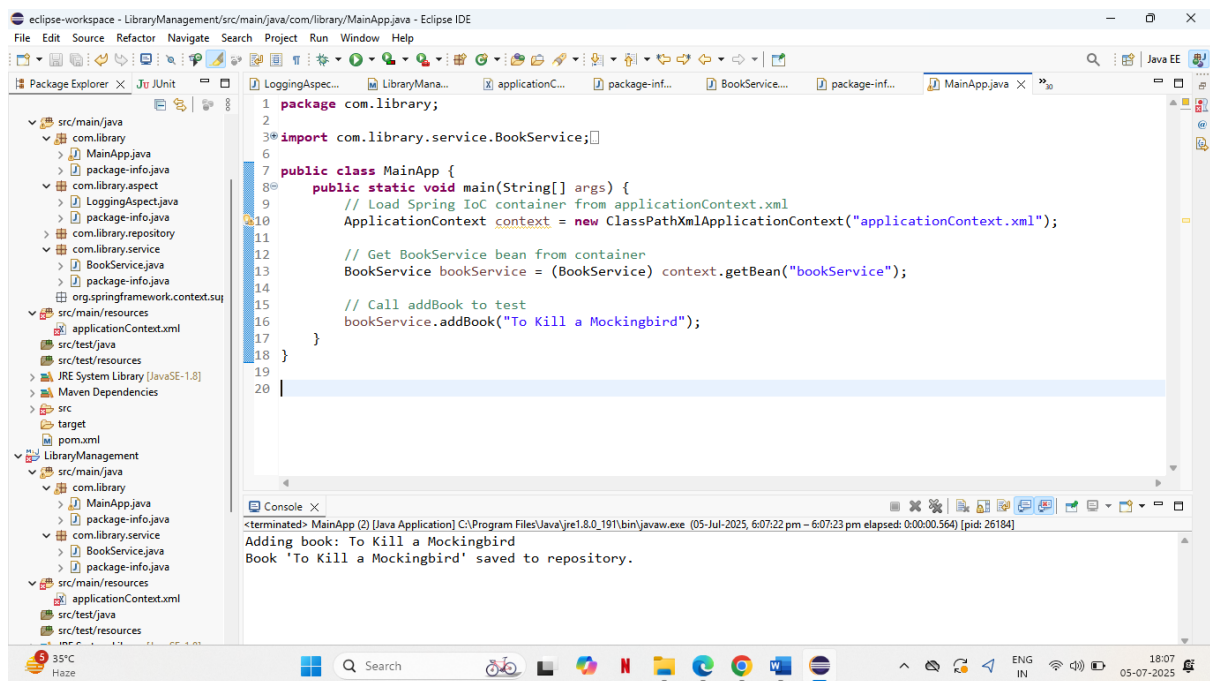
Exercise 3: Implementing Logging with Spring AOP



Exercise 4: Creating and Configuring a Maven Project



Exercise 5: Configuring the Spring IoC Container

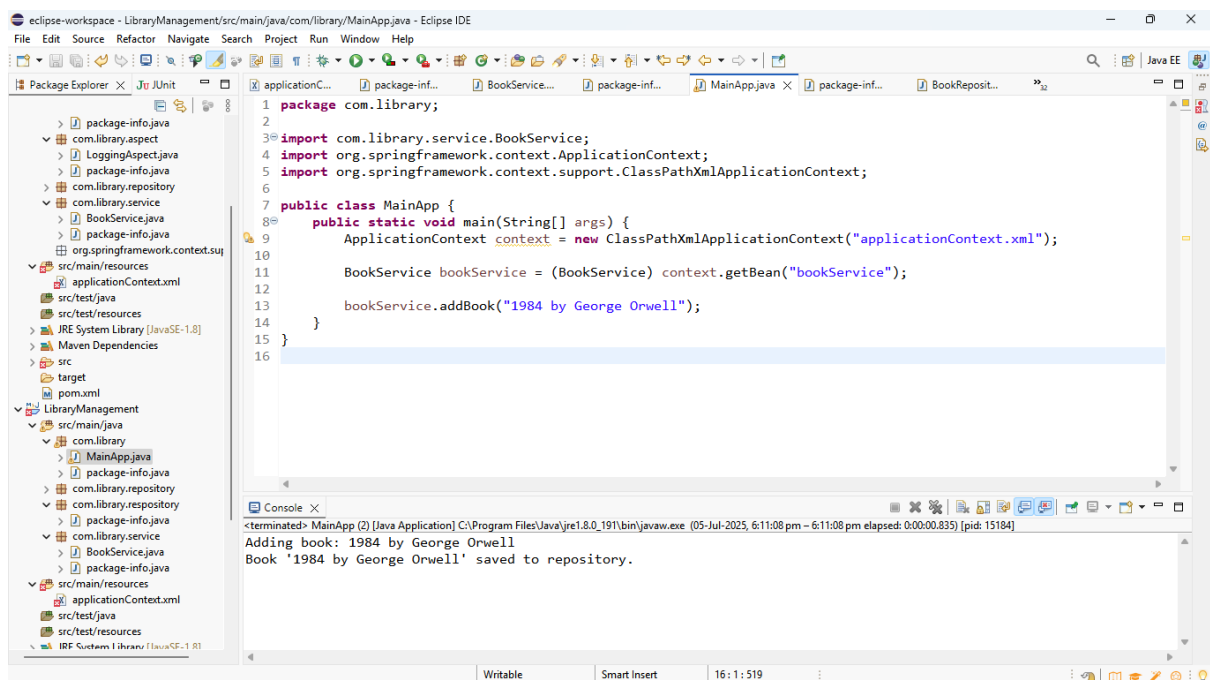


```
1 package com.library;
2
3 import com.library.service.BookService;
4
5
6
7 public class MainApp {
8     public static void main(String[] args) {
9         // Load Spring IoC container from applicationContext.xml
10        ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
11
12        // Get BookService bean from container
13        BookService bookService = (BookService) context.getBean("bookService");
14
15        // Call addBook to test
16        bookService.addBook("To Kill a Mockingbird");
17    }
18 }
19
20
```

Console Output:

```
<terminated> MainApp (2) [Java Application] C:\Program Files\Java\jre1.8.0_191\bin\javaw.exe (05-Jul-2025, 6:07:22 pm - 6:07:23 pm elapsed: 0:00:00.564) [pid: 26184]
Adding book: To Kill a Mockingbird
Book 'To Kill a Mockingbird' saved to repository.
```

Exercise 6: Configuring Beans with Annotations

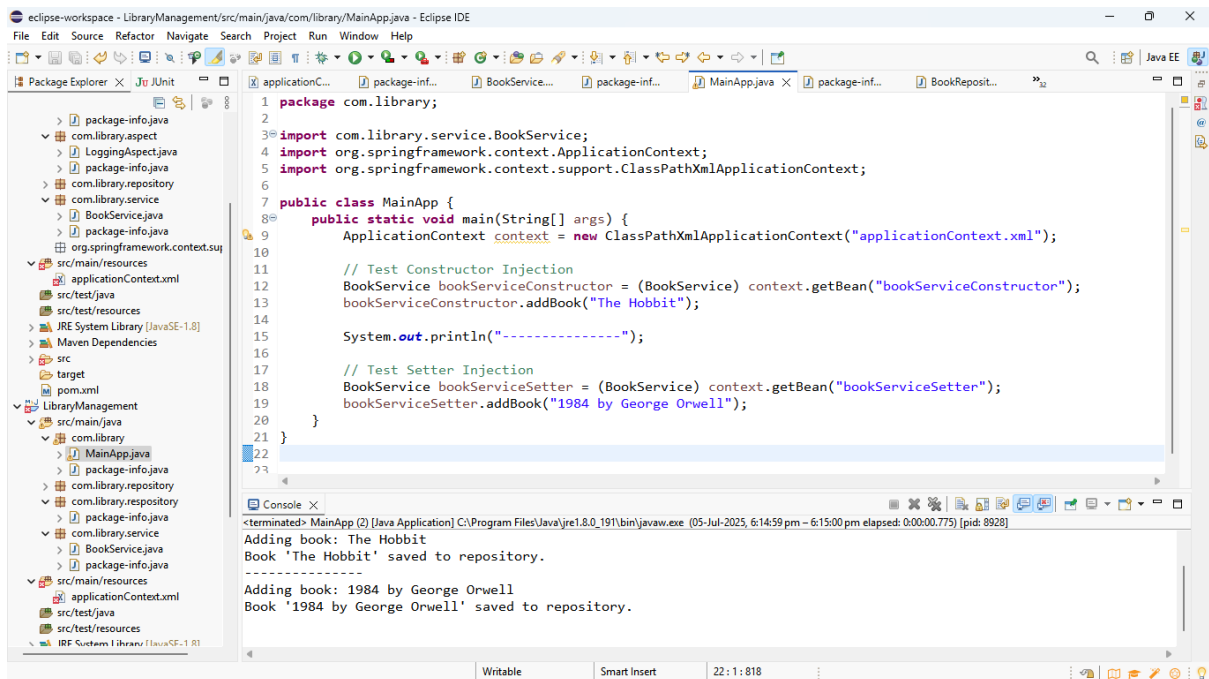


```
1 package com.library;
2
3 import com.library.service.BookService;
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6
7 public class MainApp {
8     public static void main(String[] args) {
9         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
10
11        BookService bookService = (BookService) context.getBean("bookService");
12
13        bookService.addBook("1984 by George Orwell");
14    }
15 }
16
```

Console Output:

```
<terminated> MainApp (2) [Java Application] C:\Program Files\Java\jre1.8.0_191\bin\javaw.exe (05-Jul-2025, 6:11:08 pm - 6:11:08 pm elapsed: 0:00:00.835) [pid: 15184]
Adding book: 1984 by George Orwell
Book '1984 by George Orwell' saved to repository.
```

Exercise 7: Implementing Constructor and Setter Injection

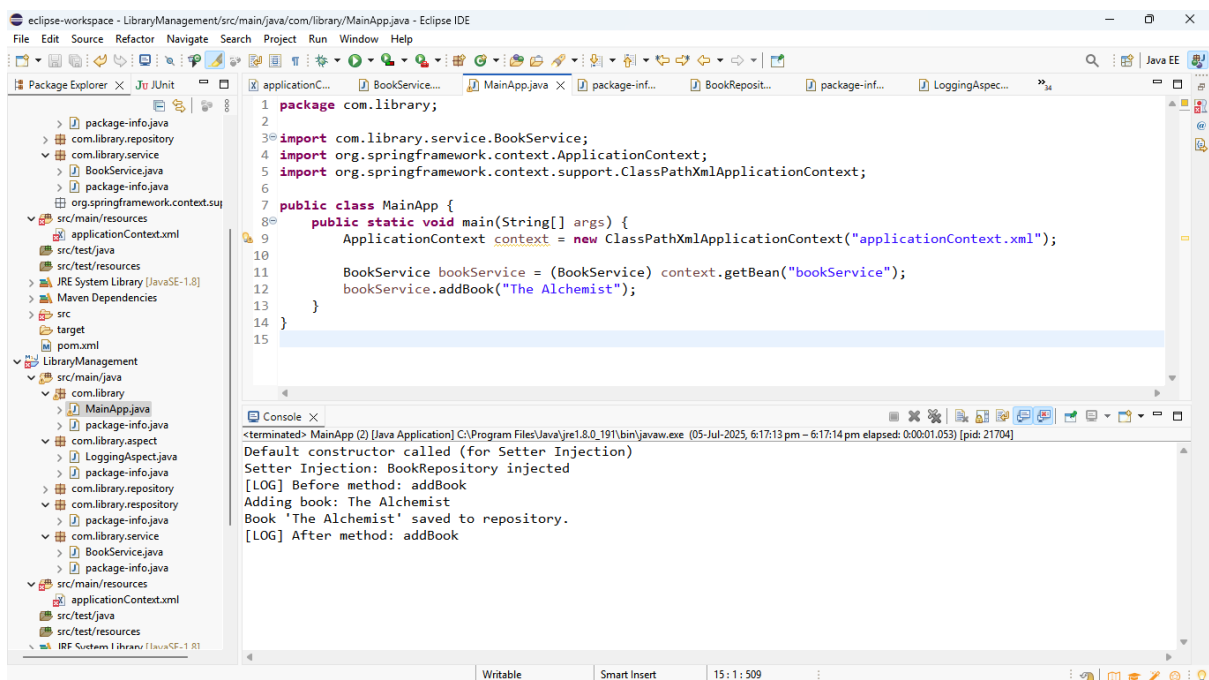


```
1 package com.library;
2
3 import com.library.service.BookService;
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6
7 public class MainApp {
8     public static void main(String[] args) {
9         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
10
11         // Test Constructor Injection
12         BookService bookServiceConstructor = (BookService) context.getBean("bookServiceConstructor");
13         bookServiceConstructor.addBook("The Hobbit");
14
15         System.out.println("-----");
16
17         // Test Setter Injection
18         BookService bookServiceSetter = (BookService) context.getBean("bookServiceSetter");
19         bookServiceSetter.addBook("1984 by George Orwell");
20     }
21 }
22
23
```

Console Output:

```
<terminated> MainApp (2) [Java Application] C:\Program Files\Java\jre1.8.0_191\bin\javaw.exe (05-Jul-2025, 6:14:59 pm - 6:15:00 pm elapsed: 0:00:00.775) [pid: 8928]
Adding book: The Hobbit
Book 'The Hobbit' saved to repository.
-----
Adding book: 1984 by George Orwell
Book '1984 by George Orwell' saved to repository.
```

Exercise 8: Implementing Basic AOP with Spring



```
1 package com.library;
2
3 import com.library.service.BookService;
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6
7 public class MainApp {
8     public static void main(String[] args) {
9         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
10
11         BookService bookService = (BookService) context.getBean("bookService");
12         bookService.addBook("The Alchemist");
13     }
14 }
15
```

Console Output:

```
<terminated> MainApp (2) [Java Application] C:\Program Files\Java\jre1.8.0_191\bin\javaw.exe (05-Jul-2025, 6:17:13 pm - 6:17:14 pm elapsed: 0:00:01.053) [pid: 21704]
Default constructor called (for Setter Injection)
Setter Injection: BookRepository injected
[LOG] Before method: addBook
Adding book: The Alchemist
Book 'The Alchemist' saved to repository.
[LOG] After method: addBook
```

Exercise 9: Creating a Spring Boot Application

The screenshot shows the Eclipse IDE interface for a Spring Boot application. The Package Explorer on the left lists the project structure, with 'LibraryManagementApplication' selected under 'src/main/java'. The main editor displays the 'pom.xml' file, which is configured for a Spring Boot project. The console at the bottom shows the build output, indicating a successful build.

```
1<?xml version='1.0' encoding='UTF-8'>
2<project xmlns="http://maven.apache.org/POM/4.0.0"
3      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
5      http://maven.apache.org/xsd/maven-4.0.0.xsd">
6
7  <modelVersion>4.0.0</modelVersion>
8
9  <groupId>com.library</groupId>
10 <artifactId>LibraryManagement</artifactId>
11 <version>0.0.1-SNAPSHOT</version>
12 <packaging>jar</packaging>
13
14 <name>LibraryManagement</name>
15 <description>Library Management System with Spring Boot</description>
16
17 <parent>
18   <groupId>org.springframework.boot</groupId>
19   <artifactId>spring-boot-starter-parent</artifactId>
20   <version>3.2.5</version> <!-- Downgraded to a stable version -->
21   <relativePath/> <!-- lookup parent from repository -->
22 </parent>
23
24 </project>
```

Console Output:

```
<terminated> C:\Program Files\Java\jdk-21\bin\javaw.exe (05-Jul-2025, 10:34:59 pm) [pid: 27300]
[INFO] Installing C:\Users\mps2k\Downloads\LibraryManagement\LibraryManagement\target\LibraryManagement-0.0.1-SNAPSHOT
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 11.605 s
[INFO] Finished at: 2025-07-05T22:35:13+05:30
[INFO]
```